

Qorvo Nearby Interaction Extended Application

Version 1.1

1 INTRODUCTION

This document is an addendum to chapter 4 “NEARBY INTERACTION DEMO APPLICATION” of the DW3000_Nearby_Interaction_QSG¹. It shows how the “Qorvo NI Demo” application for iOS works.

2 NEARBY INTERACTION EXTENDED APPLICATION

Nearby Interaction with Third Party Accessories was introduced starting from iOS-15. It is required to update an iPhone to the iOS-15 Beta version. The “Qorvo NI Demo” iOS App is available as sample code from Qorvo, instructions are provided on how to compile and run the app in a compatible iPhone device.

After install, Open the “Qorvo NI Demo” App by tapping the icon on your iOS device. The main screen shows the Connection State, Accessory State, and the Current App Status.

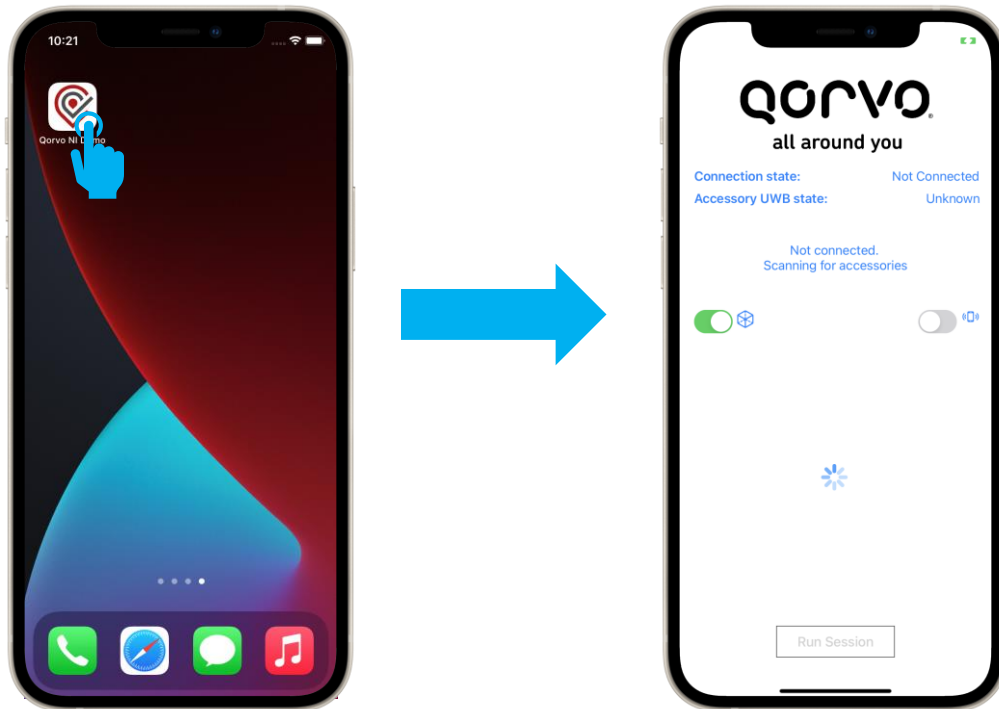


Figure 1 - "Qorvo NI Demo" App and its main screen

When “Qorvo NI Demo” opens it starts to scan for nearby accessories (Figure 1). When a Qorvo device is found it automatically connects and the “Run Session” button is enabled. When clicking on the button, the app will command the Qorvo device to start ranging.

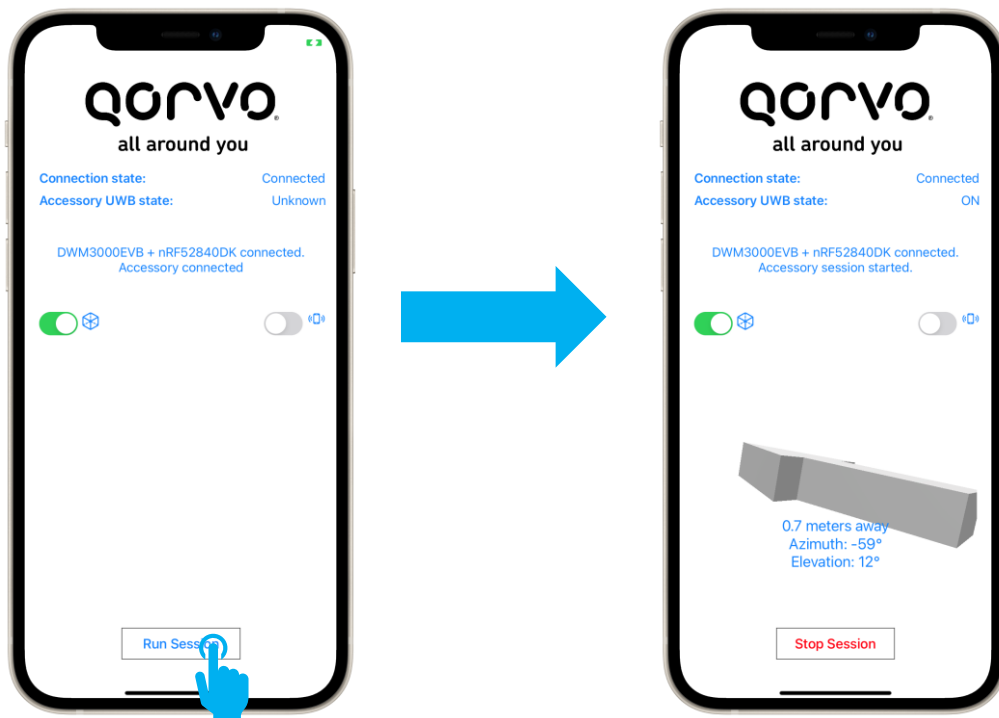


Figure 2 - Click on "Run Session" to command the Qorvo device to start ranging

After every ranging round, new position information will be available on the screen: Distance; Azimuth and Elevation.



Figure 3 - Real Time positioning information

Moving the Qorvo device towards or away from the iPhone will change the information displayed on the app (Figure 3 and Figure 4)

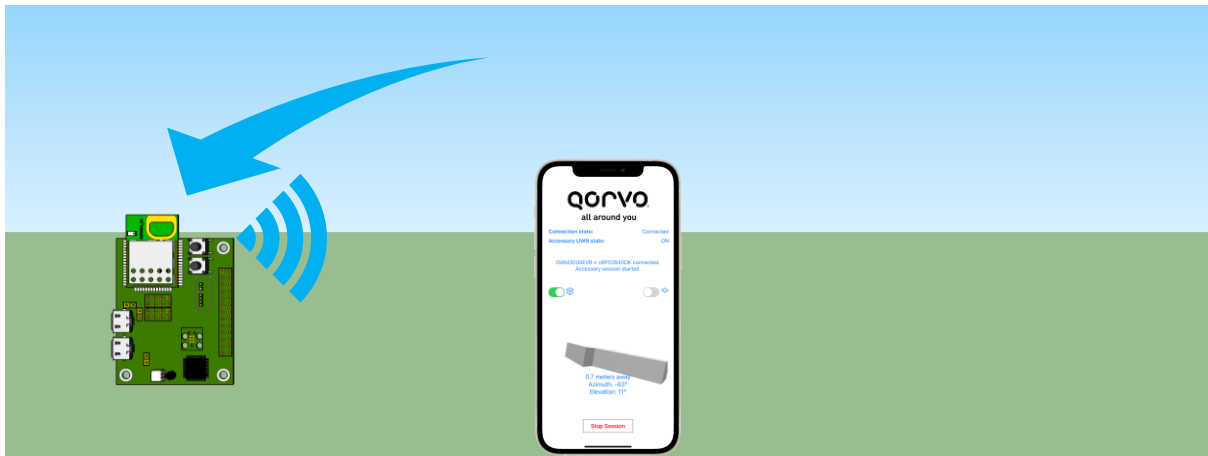


Figure 4 - Change the Qorvo device position to see the positioning info changing

The “3D Arrow” mode can be turned off by sliding the “Cube” switch. The arrow will turn in a 2D plane, using only the Azimuth angle, making it easier to visualise.



Haptic and Audio feedbacks are available for distance, by sliding the feedback switch on the main screen. Haptic feedback will use the phone vibration to report the distance varying from long, weak vibrations when far, to short and strong vibrations when close. Audio feedback will work similarly, with audio volume varying with the reported range.



Figure 5 - Enabling Haptic and Audio feedback

Feedback has three different thresholds triggered by distance: Higher than four meters, between four and two meters, and closer than two meters (Figure 6)

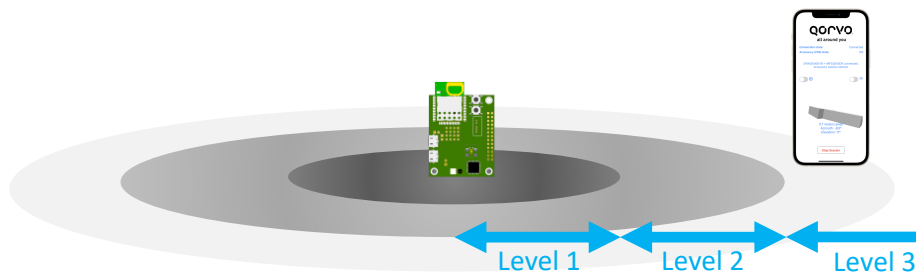


Figure 6 - Feedback levels based on Qorvo's device distance

After start ranging, the “Run Session” button changes to “Stop Session”, clicking on it will command the Qorvo device to stop ranging, the application will restart scanning for nearby Qorvo devices.

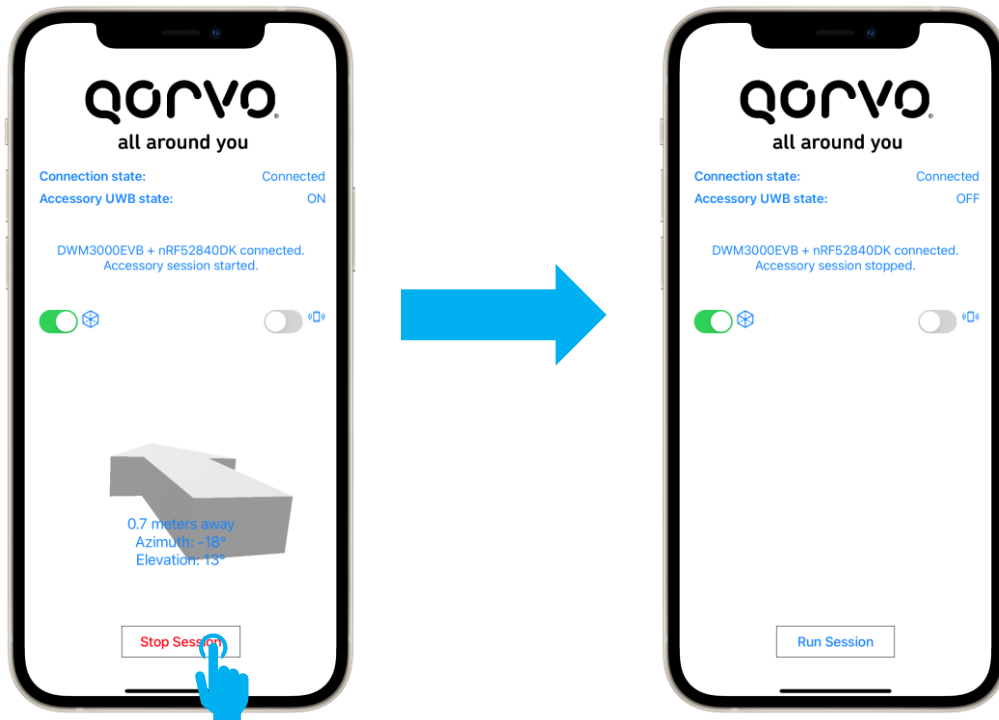


Figure 7 - Click "Stop Session" button to send the command to the Qorvo device.

3 APPENDIX 1 REFERENCES

1. DW3000_Nearby_Interaction_QSG
2. Qorvo Nearby Interaction resources
<https://www.qorvo.com/feature/uwb-solutions-compatible-with-apple-u1>